INTRODUCTION TO COMPUTER NETWORKS

WEIGHTED SUM RATE MAXIMISATION IN RSMA

SUBMITTED BY

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ABSTRACT :

The report discusses the optimization of Weighted sum rate (WSR) in Rate splitting multiple access (RSMA), a method that enhances spectral efficiency and user fairness in multi-user communication networks. RSMA achieves this by dividing each user's message into common and private streams, which helps in managing interference by using secrecy, power and privacy constraints. The study mainly aims in developing a WSR maximization approach subject to power and secrecy rate constraints, by using common algorithms like ADMM (Alternating Direction Method of Multipliers), Gradient Descent, and Adam Optimizer and Stochastic gradient descent to find the optimal algorithm. Numerical analyses and results demonstrate the comparative effectiveness of these algorithms, with Adam optimizer emerging as the most reliable for RSMA's complex requirements.

INTRODUCTION:

In modern communication systems, handling multi-user interference efficiently is critical to improving data throughput and maintaining secure transmissions. Traditional access methods like Orthogonal frequency division multiplexing and time division multiple access are limited in their ability to manage interference in densely populated networks. Rate splitting multiple access (RSMA) offers a solution by splitting messages into common streams which is decoded by all users, and private streams which is decode only by the intended recipient. This unique approach allows for improved spectral efficiency and mitigates interference across users.

This report examines a weighted sum rate (WSR) maximization problem within the RSMA framework. Specifically, it aims to allocate transmission power across users in a way that maximizes data rates while ensuring security through secrecy rate constraints. Optimization

techniques such as ADMM, Gradient Descent, Stochastic gradient descent and Adam Optimizer are tested for their efficacy in balancing power usage, secrecy, and rate allocation.

LITERATURE SURVEY:

[1] Rate-Splitting Multiple Access (RSMA) outperforms traditional methods like Space division multiple access (SDMA) and Non-orthogonal multiple access (NOMA) in downlink single-cell systems by enhancing spectral and energy efficiency. Using linearly precoded rate-splitting at the transmitter and successive interference cancellation at the receivers, RSMA can partially decode interference while treating some as noise, making it adaptable to various user deployments and network loads.

It also extends RSMA to downlink Coordinated Multi-Point (CoMP) Joint Transmission networks, optimizing beamformer design to maximize Weighted Sum-Rate (WSR) under Quality of Service (QoS) and power constraints. Numerical results confirm that RSMA significantly improves WSR over SDMA and NOMA across diverse inter-user and inter-cell channel conditions, making it a robust choice for enhanced rate and QoS in CoMP JT networks.

[2] In future multi-antenna wireless networks, efficient multiple access (MA) design is crucial to meet high throughput, diverse quality of service (QoS), and massive connectivity requirements. Traditional approaches such as Space division multiple access and non orthogonal multiple access (NOMA) rely on extreme interference management strategies—treating interference as noise (SDMA) or fully decoding and canceling interference (NOMA). In contrast, Rate-Splitting Multiple Access (RSMA) introduces a more flexible framework for downlink multi-antenna systems, which can adaptively manage interference. RSMA combines linearly precoded rate splitting with successive

interference cancellation (SIC), enabling the partial decoding of interference while treating the remaining part as noise. This hybrid approach bridges the gap between the SDMA and NOMA extremes, offering a more generalized and efficient solution.

[3] In the context of multiple-input multiple-output (MIMO) broadcast channels, the optimization of user allocation, transmit, and receive filters for maximizing sum rate is a complex, non-convex combinatorial problem. To avoid the computational burden of near-optimal implementations like Dirty Paper Coding (DPC), linear zero-forcing (ZF) approaches are commonly used. This paper introduces an efficient linear ZF algorithm that addresses this challenge by maximizing a lower bound of the sum rate. The proposed algorithm allocates data streams to users while simultaneously determining the corresponding receive filters. Compared to existing methods, the approach significantly reduces computational complexity, achieving performance improvements with minimal additional computational cost.

[4] In single-input single-output (SISO) broadcast channels, Non-Orthogonal Multiple Access (NOMA) achieves the capacity region by using successive interference cancellation (SIC) to manage inter-user interference, though SIC introduces high complexity. This paper explores Rate-Splitting Multiple Access (RSMA) as an alternative in visible light communication (VLC) networks, aiming to balance sum-rate performance with receiver complexity. The authors formulate a joint rate allocation and power control problem to maximize the sum rate while satisfying quality of service (QoS) and SIC constraints. A successive convex approximation (SCA)-based algorithm is proposed to solve this non-convex problem and obtain a local optimal solution. Numerical results show that 1-layer RSMA achieves performance close to NOMA but with significantly reduced complexity, making it an attractive alternative in SISO VLC systems.

[5] This paper explores the use of Rate-Splitting Multiple Access (RSMA) in single-input single-output (SISO) visible light communication (VLC) networks as a way to balance sum-rate performance and receiver complexity. In SISO broadcast channels, Non-Orthogonal Multiple Access (NOMA) achieves the capacity region by using successive interference cancellation (SIC) to manage inter-user interference, but SIC introduces high complexity. The authors formulate a joint rate allocation and power control problem to maximize the sum rate while satisfying quality of service (QoS) and SIC constraints. To solve this non-convex problem, a successive convex approximation (SCA)-based algorithm is proposed for a local optimal solution. Numerical results demonstrate that 1-layer RSMA provides performance close to NOMA but with significantly reduced complexity, making it an attractive alternative for SISO VLC systems.

SYSTEM MODEL:

For optimizing values in Precoding matrix 'P' and Rate Allocation Vector 'c', we perform optimization using 4 different techniques:

I. Gradient Descent.
II. Stochastic Gradient Descent (SGD).
III. Alternating Direction Method of Multipliers (ADMM).
IV. Adaptive Moment Estimation (Adam) optimizer.

These optimization techniques use the following input parameters:

- P (Precoding Matrix): It controls power allocation across antennas for each user. It is the matrix that is optimized to maximize secrecy rate while adhering to power constraints.
- c (Rate Allocation Vector): It is a vector that contains the rate corresponding to each user. It balances secrecy rate maximization and meets rate requirements.

- R_s^{th}(Secrecy Rate Threshold): It is the minimum required secrecy rate for each user for secure communication.
- N_0 (Noise Power Level): It is Signal-to-Noise Ratio(SNR).
- P_t (Total Transmit Power): It is the maximum allowed power for transmission.
- $w_{secrecy}$ (Secrecy Weight): It is weight that holds importance for secrecy rate maximization condition in the optimization process.
- α (Learning Rate): It is the learning rate that controls the step size in the gradient descent updates.

The following calculations and constrains are common across all the optimization techniques:

1. Secrecy Rate Calculation (R_s): For each user k, the secrecy rate,

$$R_{s,k} = \log_2\left(1 + \frac{\left|h_k^H \cdot P_k\right|^2}{N_0 + \text{interference}_k}\right)$$

   where:
   - $h_k$is the channel vector for user k.
   - P_k is the transmit precoder vector for user k.
   - Interference for user k is calculated as:

$$\sum_{j=1}^{k}\left|h_k^H \cdot P_j\right|^2 - \left|h_k^H \cdot P_k\right|^2$$

2. Transmit Power Constraint: It ensures that the total transmit power does not exceed a maximum allowable

$$Tr(P \cdot P^H) \leq P_t$$

   Where, Tr() denotes the trace operator.

3. Non-negativity Constraint:
   The rate allocation vector c must satisfy a non-negativity constraint

$$c \geq 0$$

4. Objective Function:
   The penalty term for user k if the secrecy rate is below $R_{th,k}$ in the Objective function,

$$\text{secrecy_penalty} = \sum_{k=1}^{K} max\left(0, R_{th,k} - R_{s,k}\right)$$

   Loss function,

$$\text{Total_loss} = \text{secrecy_penalty} + \text{penalty_factor} \times \text{secrecy_penalty}$$

The steps for updating each optimization technique are as follows:

I. <u>Gradient descent</u>:

- For each user k, the matrix P and vector c are updated as,

$$P \leftarrow P - \alpha \cdot grad_P$$
$$c \leftarrow c - \alpha \cdot grad_c$$

- The gradient grad_$R_{s,k}$ w.r.t. P_k is calculated as,

$$\text{grad_}R_\{s,k\} = \frac{2 \cdot Re(h_k^H \cdot P_k)}{ln(2) \cdot (1 + \gamma_k[k]) \cdot (\sum_{k=1}^{K} |h_k^H P_k|^2 + N_0)}$$

- The gradients grad_P and grad_c are calculated as,

$$\text{grad_}P[k] = u[k] \cdot grad_R_{s,k} + w_{\text{secrecy}} \cdot 2 \cdot grad_R_{s,k}$$
$$\text{grad_}c = u + w_{\text{secrecy}} \cdot grad_R_{s,k}$$

Gradient Descent iteratively updates P and c until the secrecy penalty is minimized and constraints are satisfied.

II. <u>SGD</u>:

For each mini-batch of users k, the gradients of $R_{s,k}$ w.r.t. P and c are calculated and updated as follows,

$$P \leftarrow P - \alpha \cdot grad_P$$

$$c \leftarrow c - \alpha \cdot grad_c$$

III. <u>ADMM:</u>

- For each user k, the matrix P and vector c are updated as,

$$P \leftarrow P - \alpha \cdot grad_P$$
$$c \leftarrow c - \alpha \cdot grad_c$$

- The gradient grad_$R_{s,k}$ w.r.t. P_k is calculated as,

$$\text{grad_}R_\{s,k\} = \frac{2 \cdot Re(h_k^H \cdot P_k)}{ln(2) \cdot (1 + \gamma_k[k]) \cdot (\sum_{k=1}^{K} |h_k^H P_k|^2 + N_0)}$$

- The gradients grad_P and grad_c are calculated as,

$$\text{grad_}P[k] = u[k] \cdot grad_R_{s,k} + w_{\text{secrecy}} \cdot 2 \cdot grad_R_{s,k}$$
$$\text{grad_}c = u + w_{\text{secrecy}} \cdot grad_R_{s,k}$$

- Updating Dual variable 'λ' and Auziliary Variable 'z',

$$z = c + \frac{\lambda}{\rho}$$
$$\lambda = \lambda + \rho(c - z)$$

Where ρ is the penalty parameter that controls the influence of the constraint c = z in the optimization process.

IV. <u>Adam optimizer:</u>

- The two moment estimates for Adam optimizer are,
  - m: the exponentially decaying average of past gradients.
  - v: the exponentially decaying average pf past squared gradients.
- For each user k, gradient of secrecy rate w.r.t. P:

$$\text{grad_}R_{s,k} = \frac{1}{ln(2)} \cdot \frac{2 \cdot Re(h_k^H P_k)}{(1 + \gamma_k|k|) \cdot (\sum_{j=1}^{K} |h_k^H P_k|^2 + N_0)}$$

- For each user k, gradients of P and c,

$$\text{grad_}P[k] = u[k] \cdot grad_R_{s,k} + w_{\text{secrecy}} \cdot 2 \cdot grad_R_{s,k}$$
$$\text{grad_}c = u + w_{\text{secrecy}} \cdot grad_R_{s,k}$$

- The Adam update for P are,

$$m_p = \beta_1 m_p + (1 - \beta_1) \cdot (grad_P)$$

$$v_p = \beta_2 v_p + (1 - \beta_2) \cdot grad_P^2$$

$$\widehat{m_p} = \frac{m_p}{1 - \beta_1^t} \qquad \& \qquad \widehat{v_p} = \frac{v_p}{1 - \beta_2^t}$$

$$P \longleftarrow P \; - \; \alpha \cdot \frac{\widehat{m_p}}{\sqrt{\hat{v}_p} + \epsilon}$$

- The Adam update for c are,

$$m_c = \beta_1 m_c + (1 - \beta_1) \cdot (grad_c)$$
$$v_c = \beta_2 v_c + (1 - \beta_2) \cdot \text{grad_c}^2$$
$$\widehat{m_c} = \frac{m_c}{1 - \beta_1^t} \qquad \& \qquad \widehat{v_c} = \frac{v_c}{1 - \beta_2^t}$$
$$c \longleftarrow c \; - \; \alpha \cdot \frac{\widehat{m_c}}{\sqrt{\hat{v}_c} + \epsilon}$$

Adam Optimizer iteratively repeats this process until convergence or until the secrecy penalty is minimized and constraints are satisfied.

RESULTS AND DISCUSSION:

In this study, we implemented and evaluated four optimization methods Gradient Descent (GD), Stochastic Gradient Descent (SGD), Alternating Direction Method of Multipliers (ADMM), and the Adam optimizer to optimize secrecy rate in a multi-user system, subject to constraints on transmit power and non-negativity for rate allocation.

1. Gradient Descent (GD):

- Secrecy Rate Penalty: 0.3264
- Transmit Power Constraint: Satisfied
- Non-Negativity of Rate Allocation: Not satisfied

2. Stochastic Gradient Descent (SGD):
    - Secrecy Rate Penalty: 2.4134
    - Transmit Power Constraint: Satisfied
    - Non-Negativity of Rate Allocation: Not satisfied
3. Alternating Direction Method of Multipliers (ADMM):
    - Secrecy Rate Penalty: 2.7673
    - Transmit Power Constraint: Satisfied
    - Non-Negativity of Rate Allocation: Not satisfied
4. Adam Optimizer:
    - Secrecy Rate Penalty: 0.8784
    - Transmit Power Constraint: Satisfied
    - Non-Negativity of Rate Allocation: Satisfied

CONCLUSION:

We have implemented and evaluated four optimization methods Gradient Descent (GD), Stochastic Gradient Descent (SGD), Alternating Direction Method of Multipliers (ADMM), and the Adam optimizer to optimize secrecy rate in a multi-user system, subject to constraints on transmit power and non-negativity for rate allocation.

Results showed that Adam outperformed the other algorithms across multiple metrics, including secrecy rate penalty, transmit power constraint, and non-negativity constraint.

Adam optimizer's adaptive learning rate and moment-based updates likely contributed to its superior performance, allowing it to navigate the optimization landscape more effectively than other methods. In contrast traditional gradient-based methods like GD and SGD were limited by

fixed learning rates and struggled to meet non-linear constraints, resulting in less stable convergence.

ADMM, while generally effective for constrained optimization, exhibited limitations in handling complex, non-convex objectives such as maximizing secrecy rate. The findings suggest that Adam's dynamic learning rate and robust update mechanism make it well-suited for secrecy rate optimization in constrained, multi-user systems, offering both high convergence stability and constraint adherence. Future research may explore hybrid or constraint-specific adaptations to further enhance Adam's suitability for secrecy rate optimization in secure communication applications.

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